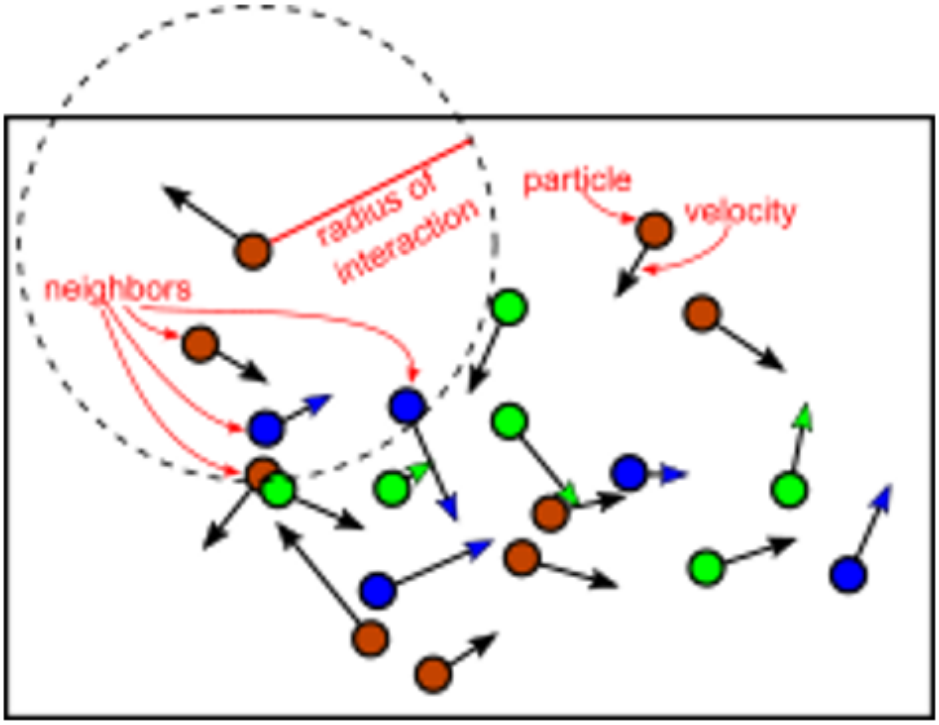



**Figure 11.10 A simplified representation of particle interaction. Several particles of different types interact in a closed environment (for instance, a gas' particles in a sealed box). Each particle has a velocity associated. For each particle, we compute the interaction only with the closest neighbors, to save computational resources.**

You see where I'm going, right? This is the perfect use case for either a N-Nearest Neighbor search, where we approximate the simulation by assuming that each particle is only influenced by the N closest particles to it (and tuning N we can trade-off precision for speed), or alternatively we can only check the interaction of each particle with the ones inside the radius of action of the four fundamental forces (or a subset of them, depending on the type of particles).